

Module 10: Essential DAX Functions

Overview

In the previous module, you learned the fundamentals of DAX, including measures, calculated columns, row context, and filter context.

This module focuses on the most important DAX functions used in real-world Power BI development. These functions form the foundation of business calculations, KPI creation, data transformation, and analytical reporting.

Mastering these functions is essential before moving to advanced DAX concepts such as `CALCULATE()`, `FILTER()`, Time Intelligence, and Iterator Functions.

Learning Objectives

After completing this module, you will be able to:

- Understand DAX function categories
 - Use mathematical functions
 - Use statistical functions
 - Apply logical functions
 - Work with text functions
 - Use date and time functions
 - Apply aggregation functions
 - Build business metrics using DAX
 - Follow DAX best practices
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1 Introduction to DAX Functions

What is a DAX Function?

A DAX function is a predefined formula that performs a specific calculation.

Syntax

```
FunctionName(argument1, argument2)
```

Example

```
Total Revenue =  
SUM(Sales[Revenue])
```

Components

```
SUM
```

Function Name

```
Sales[Revenue]
```

Argument

2 Function Categories

DAX functions are grouped into categories.

Major Categories

Mathematical Functions

Perform arithmetic operations.

Statistical Functions

Perform statistical analysis.

Aggregation Functions

Summarize data.

Logical Functions

Apply conditions.

Text Functions

Manipulate text values.

Date & Time Functions

Work with dates.

Information Functions

Provide metadata about data.

Mathematical Functions

Used for arithmetic calculations.

ABS()

Returns absolute value.

Example

```
ABS(-500)
```

Result:

```
500
```

POWER()

Raises a number to a power.

Example

```
POWER(5,2)
```

Result:

```
25
```

SQRT()

Returns square root.

Example

```
SQRT(81)
```

Result:

```
9
```

MOD()

Returns remainder.

Example

```
MOD(10,3)
```

Result:

1

ROUND()

Rounds a number.

Example

```
ROUND(25.678,2)
```

Result:

```
25.68
```



Statistical Functions

Used for statistical analysis.

AVERAGE()

Returns average value.

Example

```
Average Revenue =  
AVERAGE(Sales[Revenue])
```

MEDIAN()

Returns middle value.

Example

```
Median Revenue =  
MEDIAN(Sales[Revenue])
```

MIN()

Returns smallest value.

Example

```
Minimum Revenue =  
MIN(Sales[Revenue])
```

MAX()

Returns largest value.

Example

```
Maximum Revenue =  
MAX(Sales[Revenue])
```

STDEV.P()

Population standard deviation.

Example

```
STDEV.P(Sales[Revenue])
```

Aggregation Functions

Aggregation functions summarize data.

SUM()

Adds values.

Example

```
Total Revenue =  
SUM(Sales[Revenue])
```

COUNT()

Counts non-blank values.

Example

```
Order Count =  
COUNT(Sales[OrderID])
```

COUNTA()

Counts non-empty values.

Example

```
Customer Count =  
COUNTA(Customers[CustomerName])
```

DISTINCTCOUNT()

Counts unique values.

Example

```
Unique Customers =  
DISTINCTCOUNT(Sales[CustomerID])
```

COUNTROWS()

Counts rows in a table.

Example

```
Total Rows =  
COUNTROWS(Sales)
```

Logical Functions

Logical functions evaluate conditions.

IF()

Returns one value if condition is TRUE and another if FALSE.

Syntax

```
IF(condition,value_if_true,value_if_false)
```

Example

```
Sales Status =  
IF(  
    Sales[Revenue] > 1000,  
    "High",  
    "Low"  
)
```

SWITCH()

Alternative to multiple IF statements.

Example

```
Rating =  
SWITCH(  
    TRUE(),  
    Sales[Revenue] >= 5000,"Excellent",  
    Sales[Revenue] >= 3000,"Good",  
    "Average"  
)
```

AND()

Checks multiple conditions.

Example

```
AND(  
    Sales[Revenue]>1000,
```

```
Sales[Profit]>200  
)
```

OR()

Returns TRUE if any condition is TRUE.

Example

```
OR(  
    Sales[Revenue]>1000,  
    Sales[Profit]>200  
)
```

NOT()

Reverses logical value.

Example

```
NOT(TRUE())
```

Result:

```
FALSE
```

Text Functions

Used for string manipulation.

LEFT()

Extracts characters from left.

Example

```
LEFT("PowerBI",5)
```

Result:

```
Power
```

RIGHT()

Extracts characters from right.

Example

```
RIGHT("PowerBI",2)
```

Result:

```
BI
```

MID()

Extracts characters from middle.

Example

```
MID("PowerBI",3,4)
```

Result:

```
werB
```

LEN()

Returns text length.

Example

```
LEN("Power BI")
```

Result:

```
8
```

UPPER()

Converts to uppercase.

Example

```
UPPER("power bi")
```

Result:

```
POWER BI
```

LOWER()

Converts to lowercase.

Example

```
LOWER("POWER BI")
```

Result:

```
power bi
```

TRIM()

Removes extra spaces.

Example

```
TRIM("  Rahul  ")
```

Result:

```
Rahul
```

Date & Time Functions

Critical for business reporting.

TODAY()

Returns current date.

Example

```
TODAY()
```

NOW()

Returns current date and time.

Example

```
NOW()
```

YEAR()

Returns year.

Example

```
YEAR(Sales[OrderDate])
```

MONTH()

Returns month number.

Example

```
MONTH(Sales[OrderDate])
```

DAY()

Returns day number.

Example

```
DAY(Sales[OrderDate])
```

WEEKDAY()

Returns weekday number.

Example

```
WEEKDAY(Sales[OrderDate])
```

DATEDIFF()

Returns difference between dates.

Example

```
DATEDIFF(  
    Orders[OrderDate],  
    Orders[ShipDate],  
    DAY  
)
```



Information Functions

Provide metadata about values.

ISBLANK()

Checks if value is blank.

Example

```
ISBLANK(Sales[Revenue])
```

ISNUMBER()

Checks if value is numeric.

Example

```
ISNUMBER(Sales[Revenue])
```

ISTEXT()

Checks if value is text.

Example

```
ISTEXT(Customers[CustomerName])
```

10 Business Calculation Examples

Total Revenue

```
Total Revenue =  
SUM(Sales[Revenue])
```

Average Revenue

```
Average Revenue =  
AVERAGE(Sales[Revenue])
```

Total Profit

```
Total Profit =  
SUM(Sales[Revenue])  
-  
SUM(Sales[Cost])
```

Profit Margin %

```
Profit Margin % =  
DIVIDE(  
    [Total Profit],  
    [Total Revenue]  
)
```

Customer Count

```
Customer Count =  
DISTINCTCOUNT(Sales[CustomerID])
```



Function Nesting

Functions can be combined.

Example

```
Profit Status =  
IF(  
    [Profit Margin %] > 0.20,  
    "Good",  
    "Needs Improvement"  
)
```

Nested Example

```
Category =  
IF(  
    Sales[Revenue] > 5000,
```

```
UPPER("Premium"),  
LOWER("standard")  
)
```

1 2 DAX Best Practices

Use DIVIDE()

Instead of:

```
Revenue / Cost
```

Use:

```
DIVIDE(Revenue, Cost)
```

Use Measures Whenever Possible

Reduces memory usage.

Use Meaningful Names

Example:

```
Total Revenue
```

instead of:

```
TR
```

Keep Formulas Simple

Avoid unnecessary complexity.

1 3 Common Mistakes

Ignoring Blank Values

Can cause incorrect calculations.

Using Calculated Columns Unnecessarily

Increases model size.

Hardcoding Values

Avoid fixed numbers whenever possible.

Poor Naming Conventions

Makes maintenance difficult.



Key Terminologies

Term	Definition
Function	Predefined calculation
Argument	Input supplied to a function
Aggregation	Summarizing data
Logical Function	Evaluates conditions
Text Function	Manipulates text
Statistical Function	Performs statistical calculations
Date Function	Works with dates
Nested Function	Function inside another function
Measure	Dynamic calculation
Expression	Formula returning a value



Practice Exercises

Exercise 1

Create measures:

Total Revenue

Average Revenue

Customer Count

Exercise 2

Create a calculated column:

Revenue Category

Using IF().

Exercise 3

Extract first 3 characters from Product Name.

Exercise 4

Calculate order processing days using:

DATEDIFF()

Exercise 5

Create a Profit Status measure using:

IF()



Mini Project

Sales KPI Dashboard

Measures Required

Total Revenue

Average Revenue

Customer Count

Total Profit

Profit Margin %

Additional Tasks

Create:

- Revenue Category Column
- Profit Status Indicator
- Revenue KPI Card
- Profit KPI Card

Objective

Apply the most commonly used DAX functions in a real business reporting scenario.



Module Summary

In this module, you learned:

- ✓ Mathematical Functions
- ✓ Statistical Functions
- ✓ Aggregation Functions
- ✓ Logical Functions
- ✓ Text Functions

- ✓ Date & Time Functions
- ✓ Information Functions
- ✓ Function Nesting
- ✓ Business Calculations
- ✓ DAX Best Practices

These functions form the core toolkit of every Power BI developer and analyst. Mastering them prepares you for advanced DAX concepts, including Time Intelligence and Context Manipulation.

Next Module

Module 11: Time Intelligence in DAX

Topics Covered:

- Date Tables
- Calendar Tables
- Time Intelligence Concepts
- TOTALYTD()
- TOTALMTD()
- TOTALQTD()
- SAMEPERIODLASTYEAR()
- DATEADD()
- PARALLELPERIOD()
- Year-over-Year Analysis
- Running Totals
- Business Time-Based KPIs